Smart Public Transportation Analysis & Interactive BI Dashboard

Project Title:	Smart Public Transport Data Analytics Internship Task
Developer:	Ayesha Khan
Submission Date:	July 9, 2026
Tools Used:	Python (Jupyter Notebook, Pandas), Microsoft Excel (Pivot Tables, Slicers, KPI Formulations)

1. Executive Summary & Key Performance Indicators (KPIs)

This technical documentation outlines the end-to-end process of cleaning, transforming, and visualizing public transportation log data. Using advanced data modeling strategies, the raw data was preprocessed to resolve formatting conflicts and structured directly into an executive-level interactive Business Intelligence (BI) Dashboard.

24,920	Rs. 2,501,000	15.47 Mins
TOTAL PASSENGERS	TOTAL TICKET REVENUE	AVERAGE TRIP DELAY

2. Data Wrangling & Preprocessing Pipeline

The initial data was stored in an unformatted Comma-Separated Values (CSV) log file. A structured preprocessing workflow was executed using Python and Excel Text-to-Columns Delimiters:

- **Structural Alignment:** Split single-row comma-delimited strings into distinctive feature headers including Route_ID, Route_Name, Bus_Number, Bus_Type, Delay_Duration_Minutes, and Ticket_Revenue.
- **Data Type Enforcement:** Transformed numerical fields from text objects to fixed integers and currency metrics to avoid computation corruption during aggregation.

3. Business Intelligence Insights (Pivot Summary)

The following operational statistics were computed to evaluate route efficiency and strategic financial output:

Route Name	Calculated Route Performance (Total Revenue)
Route A (Downtown - Airport)	Rs. 572,760

Route Name	Calculated Route Performance (Total Revenue)
Route B (Saddar - Gulberg)	Rs. 459,640
Route C (Johar Town - Mall Road)	Rs. 458,220
Route D (Clifton - DHA)	Rs. 547,760
Route E (Metro Hub - Industrial Area)	Rs. 462,620
Grand Total	**Rs. 2,501,000**

4. Interactive Architecture & Dynamic Filtering

To satisfy professional submission criteria, the BI dashboard incorporates multi-layer interactive slicers linked globally across the dashboard elements:

1. **Trip Status Filter:** Allows stakeholders to dissect passenger numbers and revenue streams based on *On Time*, *Delayed*, or *Cancelled* trip metrics instantly.
2. **Bus Type Breakdown:** Isolates fleet components (e.g., Articulated Bus, Double Decker, Mini Bus, Standard Electric) to optimize route-to-vehicle allocations based on empirical high-yield demand loops.